Dr. MEENAL JAIN

+91-7838782152 | E: meenaljain.bpl1988@gmail.com

Skype-id: meenaljain.bpl1988@outlook.com

Professional Summary

Highly ambitious researcher with a background in machine learning. Expertise in data analysis and prediction. Published research papers of the same domain in reputed journals and conferences.

Skills

Feature engineering, Data visualization, Statistics, Machine Learning
Language: Python
Tools: Pandas, NumPy, PySpark
Others: Computer Networks

Projects/ Publications

Project/ Publication #1: A K-Means Clustering and SVM based Hybrid Concept Drift Detection Technique for Network Anomaly Detection.

- Construction of a hybrid Concept Drift Detection (CDD) framework for identifying drift in network traffic and improve the prediction accuracy of proposed detection model.
- Using K-Means clustering and sliding windows-based approach to reduce sample size of captured network traffic and updation of training set for model training.
- Development of two drift detection techniques, namely, Error Rate based Concept Drift Detection (ERbCDD) and Data Distribution based Concept Drift Detection (DDbCDD) for handling drift in network traffic
- An SVM based machine learning and classification to detect anomalies in the network traffic.
- Measurement and evaluation of performance of the proposed model with important metrics of accuracy, FPR, KL-Divergence, and Kappa statistics on generated testbed dataset and two publicly available popular datasets, NSL-KDD and CIDDS-2017.

Tools/ Language used: Pandas, NumPy, Python.

Published in: M. Jain, G. Kaur (Deceased), and V. Saxena, "A K-Means Clustering and SVM based Hybrid Concept Drift Detection Technique for Network Anomaly Detection," *Expert Systems with Applications*, 2022, p.116510, doi: 10.1016/j.eswa.2022.116510. (Indexing: SCI, H Index: 207, Impact Factor: 6.954)

Project/ Publication #2: A Comparison of Two Blending based Ensemble Techniques for Network Anomaly Detection in Spark Distributed Environment.

- To study distributed machine learning in Apache Spark's RDD framework concerning training time on the different number of cores,
- To study and compare blending-based ensemble models to improve the detection accuracy of anomalies in network traffic.
- The performance of the said models has been evaluated on the CIDDS dataset.

Tools/ Language used: Pandas, NumPy, PySpark, Python.

Published in: G. Kaur, and M. Jain, "A Comparison of Two Blending-Based Ensemble Techniques for Network Anomaly Detection in Spark Distributed Environment," *International Journal of Ad Hoc and Ubiquitous Computing*, vol. 35, no. 2, pp. 71-83, 2020, doi: 10.1504/IJAHUC.2020.109794. (Indexing: SCIE, H Index: 24, SJR: 0.22, Impact Factor: 0.714)

Project/ Publication #3: An ECOSVS based SVM for Network Anomaly Detection.

- Support Vector Machine (SVM) classification technique to classify normal and attack traffic in the Spark distributed environment has been introduced and evaluated.
- To reduce the cardinality of a training set an Effective COrrelation based Support Vector Selection (ECOSVS) algorithm has been proposed, which helps in reducing time and memory training complexities.
- The analysis of the said algorithm was performed on two publically available network datasets, namely, NSL-KDD

and CIDDS-2017.

Tools/ Language used: Pandas, NumPy, PySpark, Python.

Published in: M. Jain, and V. Saxena, “An ECOSVS based SVM for Network Anomaly Detection.,” *International Journal of Data Analysis Techniques and Strategies*, 2021, doi: https://doi.org/10.1504/ijdats.2022.121513. (Indexing: Scopus, H Index: 15, SJR: 0.25)

Project/ Publication #4: Distributed Anomaly Detection using Concept Drift Detection based Hybrid Ensemble Techniques in Streamed Network Data.

- To study distributed machine learning in Apache Spark’s RDD framework concerning training time on the different number of cores,
- To study the use of clustering algorithms for handling dynamic characteristics of network traffic through concept drift
- To study ensemble based SVM technique to improve the detection accuracy of anomalies in network traffic.

Tools/ Language used: Pandas, NumPy, PySpark, Python.

Published in: M. Jain, and G. Kaur, “Distributed Anomaly Detection using Concept Drift Detection based Hybrid Ensemble Techniques in Streamed Network Data,” *Cluster Computing*, 2021, doi: 10.1007/s10586-021-03249-9. (Indexing: SCIE, H Index: 50, SJR: 0.41, Impact Factor: 3.458)

Project/ Publication #5: A Study of Feature Reduction Techniques and Classification for Network Anomaly Detection.

- Reduce the dimensionality of the network traffic of recent dataset so as to lessen computational time and space complexity.
- Generate new dataset from dimensionally reduced data while maintaining the relevant features required for successful identification of new anomalies
- Apply Machine Learning Classifiers to measure performance evaluation metrics.
- Maintain or increase detection accuracy and reduce false positive rate.
- Define novel performance measures, Classification Difference Measure (CDM), Specificity Difference Measure (SPDM), Sensitivity Difference Measure (SNDM), F1 Difference Measure (F1DM) and Combined Performance Measure (CPM) to analyze the outputs.

Tools/ Language used: Pandas, NumPy, Python.

Published in: M. Jain, and G. Kaur, “A Study of Feature Reduction Techniques and Classification for Network Anomaly Detection,” *Journal of Computing and Information Technology*, vol. 27, no. 4, pp. 1-16, 2019, doi: 10.20532/cit.2019.1004591. (Indexing: Scopus, H Index: 27, SJR: 0.21)

Project/ Publication #6: A Novel Distributed Semi-Supervised Approach for Detection of Network Based Attacks.

- Optimal feature selection algorithm, namely Information Gain was used for choosing the vital attributes.
- Clustering based technique K-Means was applied to labeled the unlabeled dataset.
- These clusters were given as an input for two classification techniques namely, Decision Tree and Random Forest.
- A real time benchmark anomaly detection dataset ISCX 2012 has been used to verify the given methodology.
- Performance of the algorithms has been measured using matrices of accuracy, precision, recall, f1-score and false positive rate.

Tools/ Language used: Pandas, NumPy, PySpark, Python.

Published in: M. Jain, and G. Kaur, “A Novel Distributed Semi-Supervised Approach for Detection of Network Based Attacks,” *2019 9th International Conference on Cloud Computing, Data Science & Engineering (Confluence)*, Noida, India, 2019, pp. 120-125, doi: 10.1109/CONFLUENCE.2019.8776616.

Project/ Publication #7: A Study of Feature Reduction Techniques and Classification for Network Intrusion Detection.

- Three feature reduction techniques such as, Principal Component Analysis (PCA), Artificial Neural Network

(ANN), and Nonlinear Principal Component Analysis (NLPCA) have been used to analyze the significance of attributes.

- Three newly reduced datasets from the original benchmark dataset Coburg Intrusion Detection Data Set (CIDDS, 2017), have been created. To evaluate the performance of such reduced datasets in terms of information_ gain (IG) with actual data,
- Two metrics namely, Sensitivity_Mismatch_Measure (SMM) and Specificity_Mismatch_Measure (SPMM) have been used

Tools/ Language used: Pandas, NumPy, Python.

Published in: M. Jain, and G. Kaur, "A Study of Feature Reduction Techniques and Classification for Network Intrusion Detection," *International Journal of Advanced Intelligence Paradigms*, 2019, In press. (Indexing: Scopus, H Index: 12, SJR: 0.15)

Project/ Publication #8: Auto Associative Neural Networks for Nonlinear Principal Components Analysis of Sea Surface Temperature Anomalies in Indian Ocean.

- Optimal feature extraction algorithm, namely Principal Component Analysis (PCA) was used for choosing the vital attributes.
- Artificial neural network to study the predictability of the SST anomalies in the Indian Ocean region by using the intrinsic dimensionality of the SST time series.

Tools/ Language used: MATLAB.

Published in: M. Jain, and K. C. Tripathi, "Auto Associative Neural Networks for Nonlinear Principal Components Analysis of Sea Surface Temperature Anomalies in Indian Ocean," *Proceedings of the 9th International Conference on Advances in Soft Computing, E-Learning, Information and Communication Technology*, 20th June, 2015, PP. 58-60.

Project/ Publication #9: Machine Learning -based Model for Predicting Failure of Physical Machines in Cloud Computing.

- To study the predictability of physical machine failure in cloud computing, using machine learning model.

Tools/ Language used: Python.

Published in: M. Jain et al., "Machine Learning -based Model for Predicting Failure of Physical Machines in Cloud Computing," *Proceedings of the 3rd International Conference on Flexible Electronics on Electric Vehicles*, Springer, 28th July, 2022.

Project/ Publication #10: AdaBoost based hybrid concept drift detection approach for mental health prediction.

- Development of Fixed Sliding Windowing (FSW) algorithm for identifying drift in dataset based on warning level and drift level thresholds and improving the overall prediction of the model.

Tools/ Language used: Python.

Accepted in: M. Jain et al., "AdaBoost based hybrid concept drift detection approach for mental health prediction," *Proceedings of the 15th International Conference on Contemporary Computing (IC3-2023)*, 2023.

Patent: M. Jain, "A System for Detecting Concept Drift and Network Based Attacks in The Network Traffic," 2022, Patent Number: 202022100122 German Patent.

Work Experience

1. **Assistant Professor (Senior Grade), Jaypee Institute of Information Technology, Noida, UP |November 2022- Present**
2. **Assistant Professor, Bennett University, Greater Noida, UP | July 2022- November 2022**
3. **Assistant Professor, KIET, Ghaziabad, UP | August 2021- May 2022**

Academic Qualification

2022- Ph.D. (CSE & IT) – Jaypee Institute of Information Technology, Noida
Domain: "Application of Concept Drift and Distributed Machine Learning for Detection of Anomalies in Network Traffic".
2015 - M. Tech (Computer Science & Engineering) – IPEC, Ghaziabad- Academic Score: 72.6%
2009 - B. E (Computer Science & Engineering) – BIST, Bhopal- Academic Score: 76.2%
2005-12th (Sr. Secondary)- Academic Score: 67.3%
2003-10th (Secondary)- Academic Score: 62.8%

Extras: Certificates/Activities

1. Speaker in Faculty Development Program on "AI Tools", during June 28– July 03 (2023).
2. Professional Certificate in IBM Data Science from Coursera (July,2022)
3. National Level Workshop: A Trend towards Machine Learning: Techniques and Applications (UGC Sponsored) from Dec 26, 2017 - Jan 1, 2018, held by Department of Computer Science, Deen Dayal Upadhyaya College, University of Delhi.
4. Presented in the 9th International Conference- Cloud Computing, Data Science & Engineering (Confluence 2019) held in AMITY Noida, on 10-11 January.
5. Presented in the 9th International Conference on Advances in Soft Computing, E-Learning, Information and Communication Technology held in JNU, on 20 June.
6. Attended Faculty Development Program on Research Methodology and Tools, during August 22–26 (2022).
7. Attended Faculty Development Program on Principles of Artificial Intelligence, Machine Learning, and Deep Learning during November 26–30 (2021).
8. Attended Faculty Development Program on Performance Assessment of Computing Systems during July 10–15 (2017).

Personal Information

Name	: Meenal Jain
Date of Birth	: 15-01-1988
Father's Name	: Mr. Surendra Kumar Jain
Mother's Name	: Mrs. Sudha Jain
Nationality	: Indian
Languages	: English and Hindi
Strength	: Team Worker, Smart and Hard Worker, can adapt in any environment
Hobbies and Interests	: Web Browsing, Dancing and interacting with people

References

Prof. (Dr.) Deepak Garg
Dean & HOD
School of Computer Science and Engineering
Mobile: 9815599654
Email: deepak.garg@bennett.edu.in

Prof. (Dr.) Vineet Sharma
HOD
Dept of CSE
Mobile: 9811139463
Email: Vineet.sharma@kiet.edu

Prof. (Dr.) Vikas Saxena
HOD
Dept of CSE &IT
Mobile: 9818958936
Email: vikas.saxena@jiit.ac.in

I hereby declare that the above information about me is true and complete to the best of my knowledge.

(Meenal Jain)